PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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Project Name: KitaVisa

Version: 1.0

1. Project Overview

- **Problem Statement:** Visa-related processes in Malaysia are fragmented, manual, and heavily dependent on the user's ability to interpret unstructured information scattered across government portals, university emails, employer instructions, and verbal guidance. International students, foreign workers, and first-time applicants frequently struggle to determine the correct visa type, required documents, submission sequence, and deadlines — leading to incomplete applications, costly rejections, and significant delays. Administrative staff who support these applicants face equal burden, repeatedly answering the same procedural questions without a centralized intelligent system to offload routine decision-making.
- **Target Domain:** AI Systems and Agentic Workflow Automation — transforming unstructured, multi-step administrative visa processes into intelligent, stateful, and automated workflows powered by a General Language Model (GLM) as the central reasoning engine.
- **Proposed Solution Summary:** The AI Visa Workflow Assistant is an agentic AI system that converts unstructured visa-related user inputs — free-text queries, uploaded rejection letters, offer letters, or scanned forms — into structured, actionable, step-by-step workflows. Using Z.AI's GLM as the central reasoning engine, the system classifies intent, detects missing information, resolves ambiguity through targeted follow-up questions, and dynamically orchestrates the correct task sequence for each user's unique situation. The system shifts the user's role from manually interpreting fragmented government procedures to simply following a personalised, intelligently generated workflow — from "I don't know where to start" to "here is your next action."

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2. Background & Business Objective

- **Background of the Problem:** For years, navigating a visa application in Malaysia has followed a fragmented, manual pathway. An international student receiving a university offer letter must cross-reference the institution's immigration portal, the Immigration Department of Malaysia (JPN) website, their home country's embassy requirements, and often informal advice from seniors or administrative staff — all sources that are inconsistent, frequently outdated, and written in bureaucratic language. A foreign worker must similarly juggle employer-specific requirements, sector-specific work permit categories, and medical screening schedules with no unified guide. Every applicant effectively re-solves the same problem from scratch, and any misinterpretation — a wrong form, a missing supporting document, an expired medical report — results in rejection or delays that can last weeks to months. The process has no memory, no intelligence, and no personalisation.
- **Importance of Solving This Issue:** Automating and intelligently orchestrating this workflow reduces the cognitive burden on applicants, minimises costly errors from misinterpretation, and dramatically shortens the time between receiving an offer and completing a compliant visa submission. Beyond individual benefit, it reduces the volume of repetitive enquiries handled by university immigration units and HR departments, freeing skilled staff for higher-value tasks. For Malaysia as a destination for international students and skilled workers, a smoother, AI-assisted visa process also strengthens the country's attractiveness relative to competing destinations.
- **Strategic Fit / Impact:** This solution directly aligns with the hackathon domain requirement for AI-powered agentic workflow automation. Z.AI's GLM acts as the non-removable central reasoning engine — without it, the system cannot classify visa type, detect document gaps, resolve ambiguity, or generate personalised step sequences. The system demonstrates all four mandated GLM capabilities: understanding of unstructured inputs, multi-step reasoning and decision-making, dynamic task orchestration with tool interactions, and generation of structured actionable outputs. It is

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also strategically relevant to Malaysia's ongoing digital government transformation agenda and the growing international student population at Malaysian universities.

3. Product Purpose

3.1. Main Goal of the System: To provide an AI-powered workflow engine that transforms unstructured visa-related queries and documents into personalised, multi-step, stateful, and actionable application workflows — guiding each user from initial uncertainty to a fully prepared, compliant visa submission.

3.2. Intended Users (Target Audience):

- International students applying for Malaysian Student Passes
- Foreign workers applying for Employment Passes or Temporary Employment Visits Passes
- First-time visa applicants
- Dependent pass applicants accompanying a primary visa holder
- University international office administrative assistants (secondary users managing bulk student cases)

4. System Functionalities

4.1. Description:

The system operates as an AI-driven workflow engine. Upon receiving a user's unstructured input — whether a plain-text question, an uploaded offer letter, or a scanned rejection notice — the GLM parses intent, classifies the applicable visa category, identifies known and missing information, and generates an ordered, personalised workflow. The system maintains session state across interactions, so users can return mid-process and resume without repeating themselves. At each stage, the GLM makes a reasoning decision: proceed to the next step, request clarification, flag a risk, or escalate to a fallback. Tool integrations allow the system to

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validate document checklists, retrieve current immigration rules from the knowledge base, and surface deadline alerts.

4.2. Key Functionalities:

- Visa Intent Classification: Translates a user's free-text description of their situation — "I just got accepted to UM for my masters, what do I need?" — into a structured profile identifying visa type, institutional context, nationality considerations, and urgency level. The GLM reasons over this unstructured input to produce a classified workflow trigger.
- Stateful Workflow Orchestration: Maintains the user's workflow state across multiple sessions. Once the system knows the user is a postgraduate student applying for a Student Pass at a public university, it remembers this context and progresses through sequential stages — document collection, health screening, university endorsement, online submission — without requiring the user to re-explain their situation at each step.
- Gap Detection and Clarification: At each workflow stage, the GLM cross-references what the user has confirmed against what is required. When a required document is missing or an answer is ambiguous (e.g., "I have a letter from my university" without specifying whether it is a conditional or unconditional offer), the system pauses the workflow and surfaces targeted clarifying questions before proceeding.
- Document Parsing and Extraction: Users can upload PDF documents such as offer letters, rejection notices, or existing visa approval letters. The GLM extracts structured data — institution name, visa type referenced, dates, conditions — and automatically populates the workflow with the extracted information, reducing manual entry and eliminating common transcription errors.

4.3. AI Model & Prompt Design

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4.3.1. Model Selection: The system uses Z.AI's GLM as its central reasoning engine. GLM is selected for three specific capabilities that are non-negotiable for this use case. First, its strong performance on long-context, multi-turn reasoning makes it suitable for maintaining stateful workflows across extended user sessions without losing track of previously established facts. Second, GLM's ability to handle mixed input types — free text, structured form data, and document-extracted content — allows the system to unify the fragmented input landscape that visa applicants face. Third, GLM's instruction-following accuracy under a structured system prompt with agentic decision logic enables the reliable step-by-step orchestration that the workflow engine requires. A simpler model that cannot reason about ambiguity or multi-step dependencies would produce a chatbot, not a workflow engine — which is precisely the limitation this product is designed to overcome..

4.3.2. Prompting Strategy: The system uses a multi-step agentic prompting strategy. A master system prompt establishes the GLM's role as a visa workflow orchestrator and defines its decision logic: classify intent, check state, detect gaps, generate next action, and handle failures. Individual workflow stages are managed through chained prompts where each GLM call receives the current session state — visa type, completed steps, confirmed documents, open gaps — as structured context, and returns a structured JSON response specifying the next action, any clarifying questions, and an updated state object. This agentic approach is chosen over zero-shot or few-shot prompting because visa workflows are inherently conditional and stateful — the correct next step depends on accumulated context across many turns, not just the most recent user message. Few-shot examples are embedded within stage-specific prompt templates to guide output format consistency.

4.3.3. Context & Input Handling: The system accepts three input types: free-text queries (up to 2,000 tokens per turn), uploaded documents (PDF up to 5MB, extracted to text before being passed to the GLM), and structured form field responses. For uploaded documents, text is extracted and truncated to the most relevant 3,000 tokens using a pre-processing step that prioritises content near detected keywords (visa type names, institution names, dates, document titles). Inputs exceeding the per-turn limit are chunked by paragraph boundary and processed sequentially, with extracted entities consolidated before being passed to the orchestration layer.

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Inputs that are entirely unreadable (e.g., corrupted PDFs or image-only scans without OCR content) are rejected gracefully with a user-facing message requesting a re-upload or manual entry of the key details.

4.3.4. Fallback & Failure Behavior: When the GLM returns a response that fails structured output validation — missing required fields in the JSON workflow state, off-topic content unrelated to the current workflow stage, or an internally inconsistent next-step recommendation — the system triggers an automatic retry with an appended correction instruction in the prompt (up to two retries). If both retries fail, the system enters a graceful degraded state: it preserves the last valid workflow state, informs the user that it encountered difficulty processing the request, and offers three options — rephrase the input, skip the current step and return to it later, or contact a human support officer. Hallucination risk is mitigated by grounding all immigration rule references in a curated internal knowledge base rather than allowing the GLM to generate procedural facts from parametric memory alone. Any GLM output referencing a specific document requirement or deadline is cross-validated against the knowledge base before being surfaced to the user.

5. User Stories & Use Cases

- **User Stories:**
  - As an international student who just received a university offer letter, I want to upload my offer letter and immediately receive a personalised checklist of exactly what I need to apply for a Student Pass, so that I do not have to manually read through JPN guidelines and risk missing a required document.
  - As a foreign worker whose Employment Pass application was rejected, I want to paste the rejection email into the system and receive a clear explanation of which condition was unmet and what corrective steps I must take, so that I can resubmit correctly without paying a consultant.

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- As a university international office administrator, I want to track the workflow progress of multiple student cases simultaneously, so that I can proactively follow up with students who are stalled at a particular stage before their deadlines expire.
- As a dependent pass applicant, I want to answer a short set of guided questions about my spouse's visa status and receive a step-by-step action plan, so that I do not accidentally apply under the wrong visa category.

- **Use Cases (Main Interactions):**
  - **Unstructured Query to Workflow**: The user types a free-text description of their situation. The GLM classifies visa type, identifies nationality context, and generates a personalised multi-step workflow displayed as an ordered task list with estimated timelines. The user's session is initialised and state is saved.
  - **Document Upload and Auto-Population:** The user uploads an offer letter or rejection notice. The system extracts key entities (institution, visa type referenced, dates, conditions stated), pre-fills the relevant workflow fields, highlights what was successfully extracted, and asks the user to confirm or correct before proceeding. This eliminates manual re-entry and reduces transcription errors.
  - **Gap Detection and Clarification Interruption**: At any workflow stage, if the GLM detects that a required input is missing or ambiguous, it pauses forward progression and presents a focused clarification prompt — one or two specific questions — before unlocking the next step. The user cannot skip a mandatory gap without either answering the question or explicitly flagging that the information is unavailable, which triggers an alternative pathway.
  - **Failure Recovery and Escalation:** If the user provides input the system cannot process reliably, the system presents a clear error state, preserves all prior workflow progress, and offers the option to contact a human support officer with a pre-filled summary of the user's current workflow state to avoid repeating information.

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- **Use Case Diagram:** *(Optional but encouraged for participants to visualize the interaction)*.

6. Features Included (Scope Definition)

- Conversational, multi-turn visa workflow interface for free-text query input and guided question-and-answer sessions
- Document upload and GLM-powered extraction for PDF offer letters, rejection notices, and existing visa approval documents
- Stateful session management that persists workflow progress across multiple logins
- Personalised step-by-step action plan generation, specific to visa type, institution type, and nationality
- Gap detection engine that identifies missing documents or ambiguous inputs and surfaces targeted clarifying questions
- Curated immigration knowledge base covering Student Pass, Employment Pass, and Dependent Pass procedures in Malaysia
- Graceful failure and fallback states with human escalation pathway and pre-filled context handoff

7. Features Not Included (Scope Control)

- Direct integration with JPN (Immigration Department of Malaysia) or university portals for automated form submission — the system guides and prepares, but does not submit on the user's behalf
- Real-time visa status tracking via government API (not publicly available at this time)
- Support for visa categories outside Student Pass, Employment Pass, and Dependent Pass in the initial version
- Multi-language support beyond English and Bahasa Malaysia in the initial version
- Mobile native application — web interface only in the initial version
- AI-generated legal advice or guarantees of application success — all outputs are informational guidance only

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8. Assumptions & Constraints

- **LLM Cost Constraint:** An average user session is estimated to involve 8–12 GLM calls across a complete workflow (intent classification, per-stage orchestration, gap detection, and output generation), consuming approximately 6,000–10,000 tokens per session. To keep inference costs manageable at scale, the system caches visa rule lookups from the knowledge base so that repeated queries for the same visa category do not trigger redundant GLM calls. Additionally, document extraction pre-processing is handled by a lighter rule-based parser before GLM is invoked, reducing the token volume passed to the model for routine field extraction.

- **Technical Constraints:** The document parser supports PDF and plain text inputs only. Image-only PDFs (scanned documents without embedded text) require OCR pre-processing, which is not guaranteed to be available in all deployment environments during the hackathon build. The knowledge base covers Malaysian immigration rules as of the current version and requires manual updating when JPN policies change — there is no live government data feed.

- **Performance Constraints:** Workflow generation for complex cases involving multiple visa categories or ambiguous inputs may require up to three sequential GLM calls before a complete plan is produced, introducing latency of several seconds. Users are shown a progress indicator during processing. To avoid excessive token usage, the maximum number of active workflow stages displayed simultaneously is capped at ten; completed stages are archived but remain accessible.

- **User Input:** The GLM cannot initiate a workflow without at least a minimal user input describing their situation. The system is not designed for production immigration legal use and all outputs carry a disclaimer that users should verify current requirements directly with their institution or JPN before submitting any application.

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9. Risks & Questions Throughout Development

- **Workflow Accuracy vs. Policy Drift:** How do we ensure the GLM-generated workflow remains accurate as JPN immigration policies update — sometimes without advance notice? The risk is that a user follows a step sequence that was correct at build time but has since changed. Mitigation: the knowledge base is version-controlled and dated, and all workflow outputs include a notice to verify requirements against the official JPN portal before submission.

- **Ambiguity Misclassification:** What happens if the GLM incorrectly classifies a user's visa type at the intent classification stage, generating an entirely wrong workflow? This is the highest-impact failure mode. Mitigation: the system always displays the classified visa type prominently at the start of the workflow and requires explicit user confirmation before proceeding. Users can correct the classification at any point, which triggers a workflow reset.

- **Hallucinated Document Requirements:** What if the GLM surfaces a document requirement that does not actually exist, causing the user to spend time obtaining a non-existent form? Mitigation: all document requirements surfaced to the user are cross-validated against the curated knowledge base. GLM is explicitly instructed not to generate document names or requirements from parametric memory alone.

- **Incomplete Document Extraction:** What happens when an uploaded offer letter is poorly formatted or uses non-standard phrasing, causing the extractor to miss key fields? Mitigation: any field that cannot be extracted with sufficient confidence is flagged to the user for manual confirmation rather than being silently assumed. The system never populates a critical field (visa type, institution name, dates) without user verification.

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